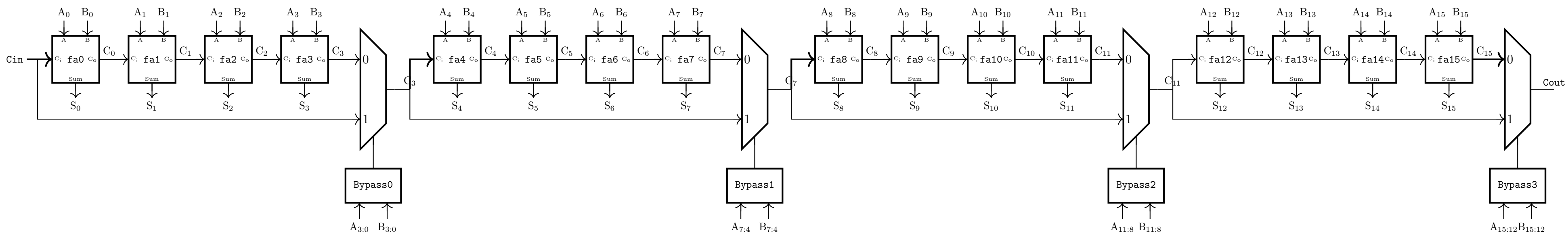
EE5311 Digital IC Design: Assignment 8
Carry bypass adder – schematic and layout extracted simulation

1. Schematics and symbols for `full adder, xor, mux2_1, nand4, inverter` gates are in `/cad/share/tutorial_8` directory. Copy over these files to your working directory. Construct a 16-bit carry bypass adder using these gates as shown below:



   (a) Trace the critical path in the above schematic and come up with an input pattern that activates the critical path.

   (b) Create a SPICE stimulus for the pattern and measure the critical path delay in `ngspice`.

2. Firstly run the `update_std_cells.sh` command to get the required standard cell layouts in KLayout. Close and Open KLayout to get the new standard cells. Draw the layout for the 16-bit adder using the cell layouts provided in the updated `custom_sky130` library.

   (a) Ensure that the layout is LVS/DRC clean

   (b) Measure the post-layout extracted delay of the critical path